CHAI: Fairness

Linguistics of AI Ethics Charters & Manifestos

Project Report

Project Supervisor: Maria Boritchev

Team: Chengbo Zhang, Constance Nguyen, Hugo Fanchini, Mourad Latoundji

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I. Introduction

In recent years, the rapid growth of artificial intelligence has been accompanied by a proliferation of initiatives seeking to articulate what ethical AI should entail. These efforts have produced a diverse set of charters and manifestos authored by academic institutions, industry actors, non-profit organizations, regulatory bodies, and civil society groups. Their formats range from concise normative statements to comprehensive assessments or records of public deliberation. Collectively, these documents form a heterogeneous but valuable resource for understanding how different stakeholders conceptualize the values, priorities, and tensions surrounding AI ethics (Mapping AI Ethics: a meso-scale analysis of its charters and manifestos, M. Gornet et al., 2024).

The present project investigates the linguistic properties of this corpus of AI ethics charters and manifestos, referred to as MapAIE. We focus on the semantic and structural behavior of a single key term — **fairness** — and study how its meaning and usage vary across the corpus. To do so, we combine classical lexical analysis with the Abstract Meaning Representation (AMR) framework, which offers a structured view of sentence-level meaning.

In addition to the analytical framework, we developed a dedicated processing pipeline to support data cleaning, AMR segmentation, and fairness-focused analysis. All scripts and tools used throughout the project are openly available in a companion GitHub repository (https://github.com/Hugosh71/Projet_CHAI_Fairness_TelecomParis), ensuring transparency and reproducibility of our methodology.

Our work is organized into three main components:

- Data retrieval and exploration
- Analysis of the occurrences and contextual behavior of the term *fairness*
- AMR-based semantic exploration of how fairness is positioned and interpreted within individual sentences.

This combined approach provides insight into how fairness is invoked, conceptualized, and integrated into broader ethical discourse across diverse institutional sources, while the accompanying codebase enables full reproducibility of the analysis conducted.

II. Data Retrieval and Exploration

II.1. Mapping AI Ethics (MapAIE) Dataset

The *Mapping AI Ethics (MapAIE)* dataset, developed by researchers at Télécom Paris (Institut Polytechnique de Paris), compiles and standardizes AI ethics charters and manifestos from various institutions worldwide. It was created to address the lack of a unified format for analyzing such documents and is primarily intended for textual and structural analysis.

The dataset contains 730 documents (of which 436 are included in the current version), each annotated with attributes such as title, institution, authors, sector, country, date, and inclusion status. The data is drawn from freely available online documents discussing artificial intelligence and ethics, written in English, and filtered using strict inclusion criteria.

The collection process involved manual human annotation and automated web scraping. Only preprocessed text (not full documents) is released to comply with copyright and licensing restrictions.

The dataset has been used for exploratory analysis and topic modeling and can serve other natural language processing (NLP) purposes, for example, keyword-based exploration.

II.2. Data Retrieval and Completeness Problem

The MapAIE project provides an open-source framework for constructing and analyzing the *Mapping AI Ethics* dataset. The corresponding codebase is publicly available at https://gitlab.telecom-paris.fr/tiphaine.viard/mapaie. This repository enables researchers to reconstruct the MapAIE dataset and reproduce the analysis presented in the associated research.

In our work, we used this repository to build the dataset used in our study. We primarily relied on two key components of the MapAIE codebase:

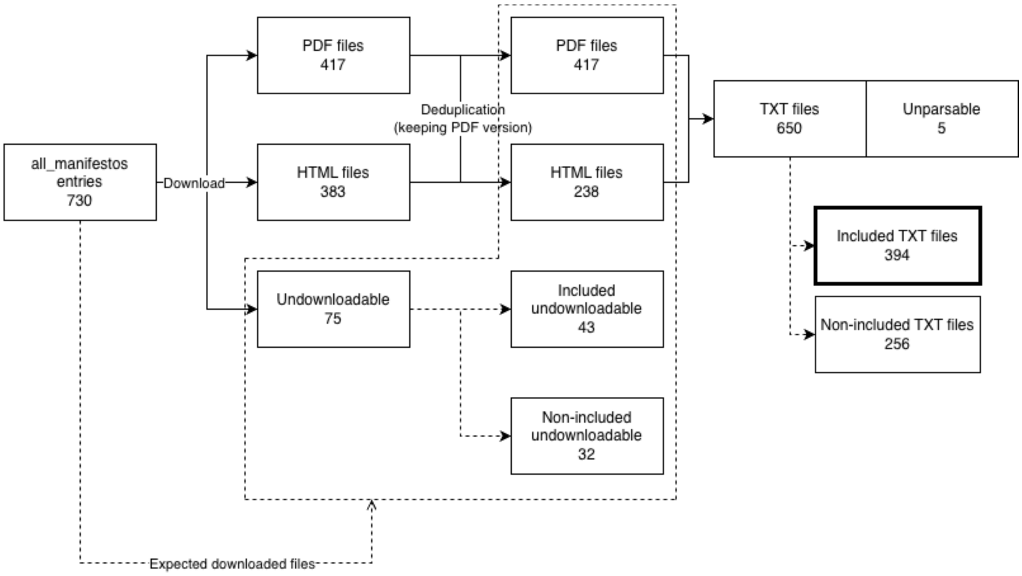
1. Tools for document retrieval and parsing, including the scripts:
   a. dl_docs.py – for automated downloading of source documents;
   b. parse_docs.py – for parsing and extracting textual content from HTML and PDF files.
2. A manifest of AI ethics documents in CSV format, named all_manifestos.csv, which lists all known AI ethics charters and manifestos, along with metadata such as title, institution, URL, authors, country, sector, publication date, and inclusion/exclusion criteria.

Each entry in the manifest corresponds to one potential document instance. An example excerpt is shown below:

Name of the document	Institution	URL	Authors	Sector	Country	Date	Status	Label
A Unified Framework of Five Principles for AI in Society	–	...	–	–	–	–	Not included	meta analysis
AI Decolonial Manyfesto	–	...	Aarathi Krishnan et al.	Civil society	International	2021	Not included	not cited

Intel Recommends Public Policy Principles for Artificial Intelligence	Intel	...	Naveen Rao	Industry	USA	–	Included	SPI

We executed the scripts to download the PDF and HTML documents from the URLs listed in all_manifestos.csv, and then parsed them into TXT files, as summarized below:



From the 730 entries listed in *all_manifestos.csv*, a total of 800 documents were successfully downloaded — comprising 417 PDF files and 383 HTML files. After deduplication (retaining the PDF version when duplicates existed), the dataset contained 655 unique documents.

Among these, 5 documents could not be parsed due to issues such as corrupted content or invalid file formats (raising the *"NOT A RECOGNIZED FILETYPE"* error). These files were excluded to maintain data consistency and integrity.

Following parsing, 650 TXT files were produced. Filtering by "Included" status yielded 394 included TXT files and 256 non-included TXT files, where the "Included" annotation marks entries that are relevant to AI ethics and written in English.

Because both the MapAIE paper and the associated repository were published in 2024, some online sources have since become inaccessible, leading to *HTTP 403* (forbidden) or *404* (not found) errors. Consequently, several documents labeled as *"included"* could not be retrieved.

Examples of *included but undownloadable* documents:

Name of the document	**Status**	**Label**	**HTTP Code**
Artificial intelligence and the control of COVID-19	Included	SPI, issue-specific	403
Artificial Intelligence: A European Perspective	Included	SPI	404
Guidance on AI and Data Protection	Included	Issue-specific	404
G20 AI Principles	Included	–	403
AI Policy Principles	Included	SPI	404
Malta, the Ultimate AI Launchpad: A Strategy and Vision for AI in Malta 2030	Included	SPI	403

Culture, Platforms and Machines: The Impact of AI on the Diversity of Cultural Expressions	Included	SPI, issue-specific	404

After the parsing and filtering process, we kept a total of **394 included text files**. Among the extracted text files, several still contained residual CSS code that needed to be removed. This cleaning step was performed using the *remove_css* command within *preprocess.py* in our repository.

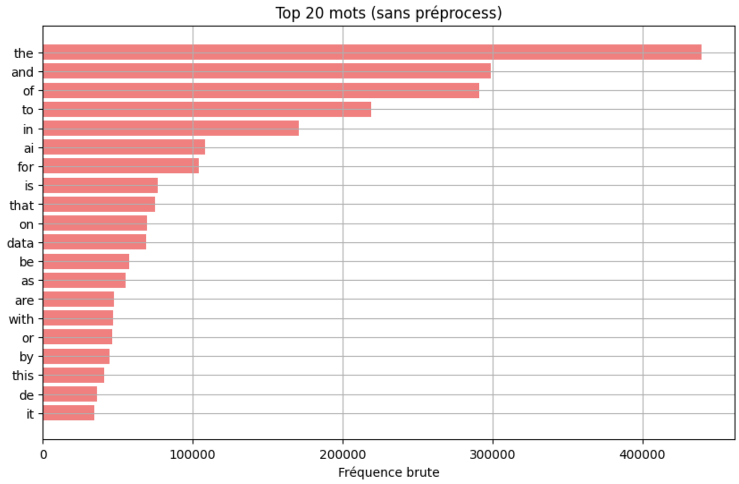
The conversion from the original heterogeneous sources PDF and HTML into plain text format wasn't an easy step in ensuring both homogeneity and usability of the dataset. This transformation necessarily implied the loss of visual structure, typographical emphasis and certain contextual information but it facilitated large-scale quantitative analysis, tokenization and word frequency computation.

II.3. Data Exploration

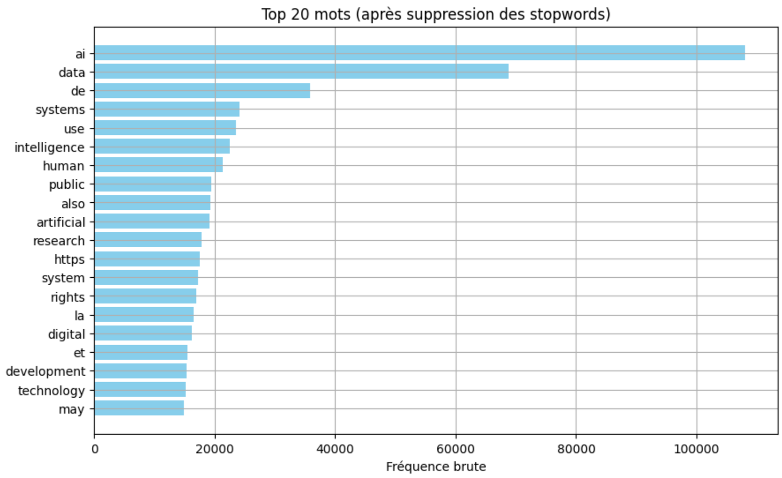
Once the corpus was prepared, the first step of analysis consisted in exploring the overall lexical composition by identifying the most frequent words, excluding stopwords, in order to get an "initial" understanding of the dominant vocabulary/words used in AI ethics discourse and situates the notion of fairness within a semantic field.

Three complementary approaches were implemented:

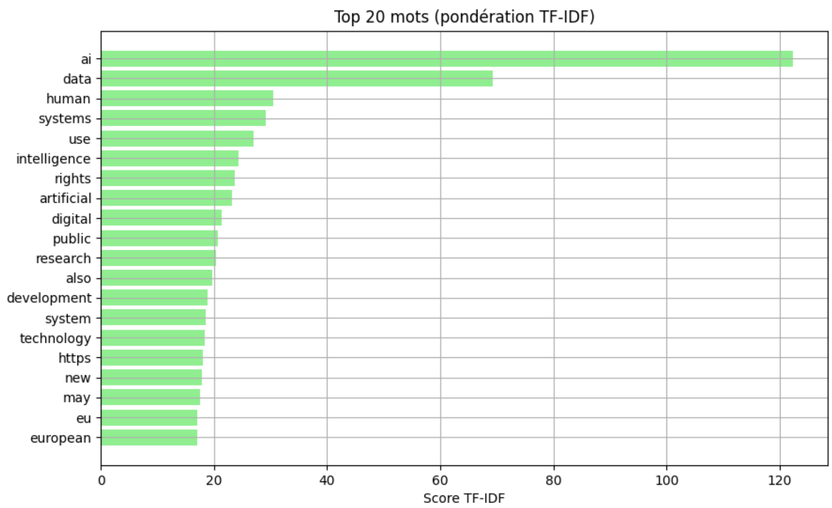
- A first analysis of raw word frequencies without preprocessing



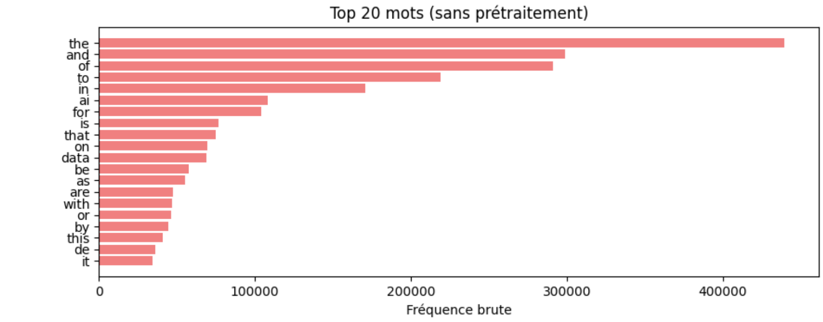
- A second analysis of word frequencies, but this time after the systematic removal of stopwords, which revealed the centrality of terms such as ai, artificial, intelligence, ethics, data etc. Those terms reflect the core themes addressed across manifestos.

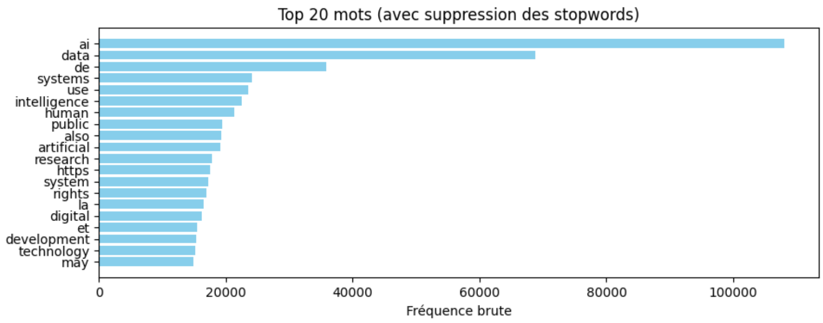


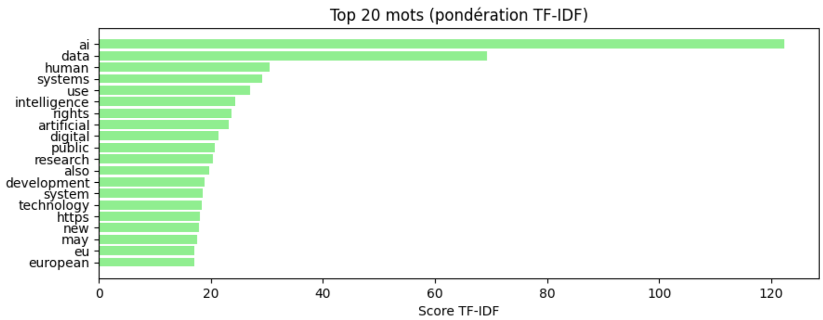
- A third analysis, this time employing the term frequency-inverse document frequency (TF-IDF), weighting scheme to identify terms that are distinctive within specific documents rather than merely frequent across the corpus.



Overall comparison between the three approaches:

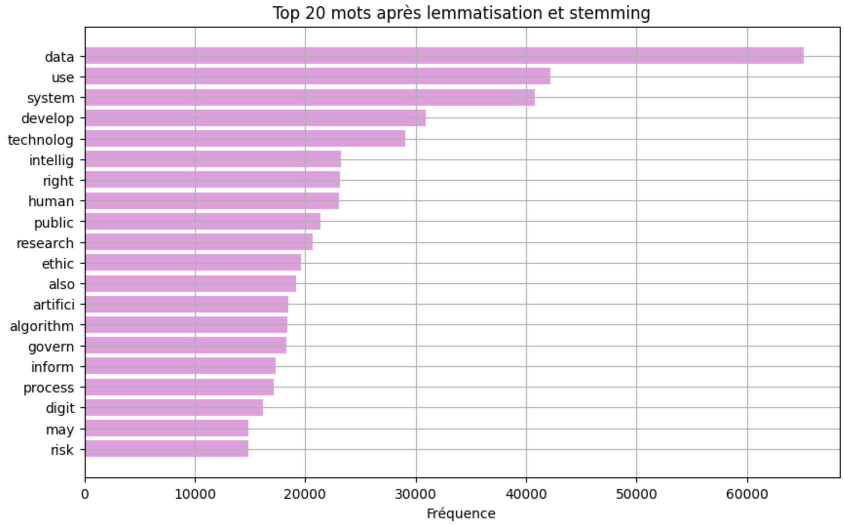




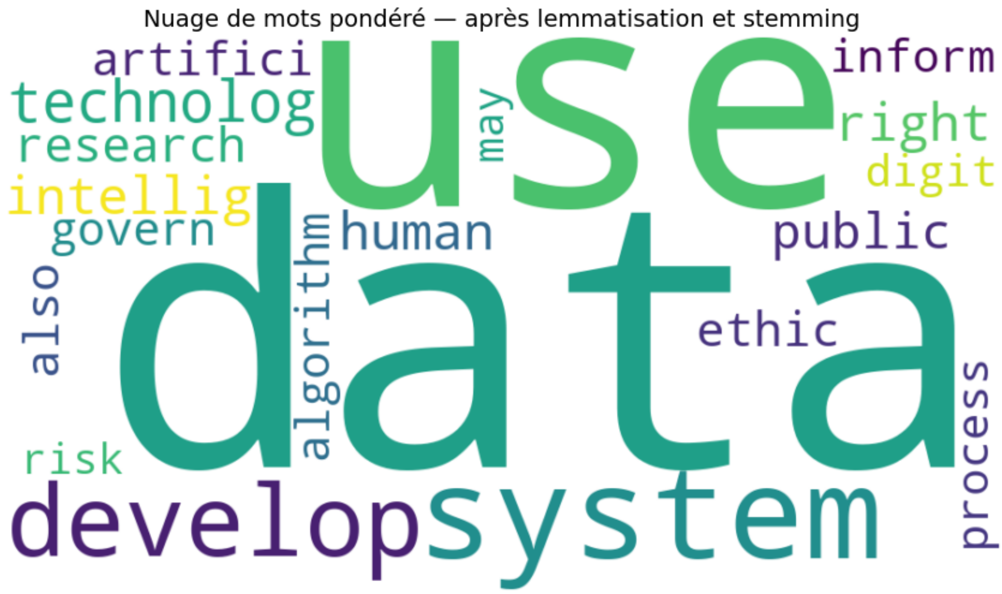


The results indicate that the corpus is not only concerned with abstract ethical principles but also with the operationalization of these principles in the governance of AI systems. The persistence and overall presence of the word fairness within the corpus confirm its central position in the discourse and justify the focus of the subsequent semantic analysis.

To go further, we applied lemmatization and stemming techniques to reduce morphological variation and lexical redundancy. Lemmatization allowed the consolidation of word families (e.g., tech, technology, technological) under a "unified" form, while stemming provided a complementary verification of dimensionality reduction. We then established another ranking based on words after lemmatization and stemming.

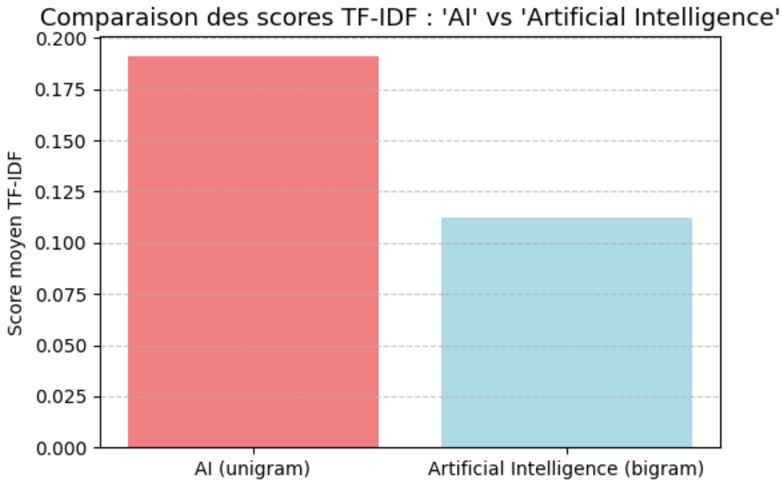


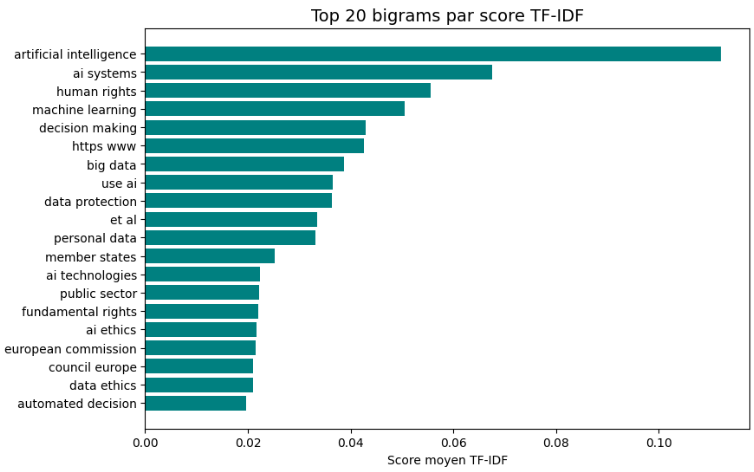
We concluded this part by generating a word cloud from the lemmatized corpus to get a visualization of these trends. The cloud highlights the predominance of certain notions already mentioned in previous graphics such as “data”, “develop”, “technolog” or “artifici”.



Following the exploration of word frequencies, we extended the analysis to multi-word expressions (n-grams) to capture recurring conceptual units that might be lost when analyzing isolated words. We gave particular attention to the bigram “artificial intelligence” and the unigram “ai”, which obviously both refers to the same concept but may differ in usage and distribution across the corpus. The objective was to determine whether these two lexical forms are used interchangeably or if they carry different contextual meanings.

To do that, we used a TF-IDF analysis on bigrams extracted from the corpus, with the same preprocessing pipeline as the previous analysis (tokenization, stopwords removal, etc.).





The TF-IDF comparison between AI and artificial intelligence reveals a clear distinction in their distribution across the corpus. We can see that AI obtains a higher average TF-IDF score, suggesting that it appears more frequently as a distinctive and contextually salient term across individual documents.

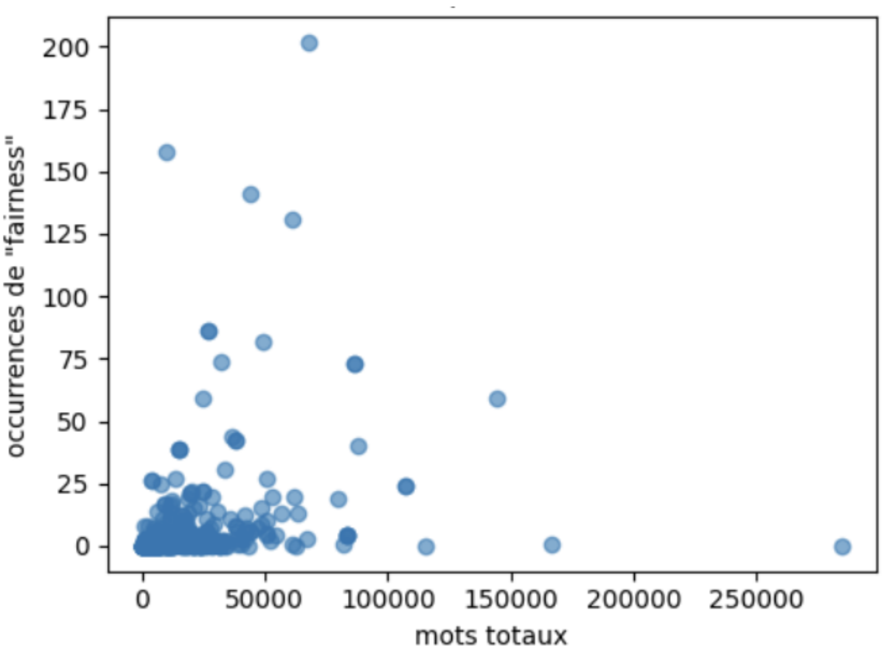
This difference can be explained by the way each expression functions linguistically. The shorter form AI is often used in policy summaries, industrial charters or reports where concision is needed. Meanwhile, the full bigram artificial intelligence can appear in a more formal context like publication titles.

Overall, both diagrams suggest that while AI is more context-specific according to the TF-IDF, and artificial intelligence continues to play a central role as the core phrase around which the ethical discourse is constructed.

III. Exploration of the Term “Fairness” in the Data

III.1. Frequency and occurrences of the Term “Fairness”

We load raw texts from the MapAIE corpus and tokenize words using a light Unicode-aware regex to handle accents and hyphens. For each document we compute: the raw count of the token “fairness”, total word count, and a length-normalized rate (occurrences per 10k words). Visualizations are built with matplotlib (scatter plots of raw vs. normalized rates; bar charts for top documents).



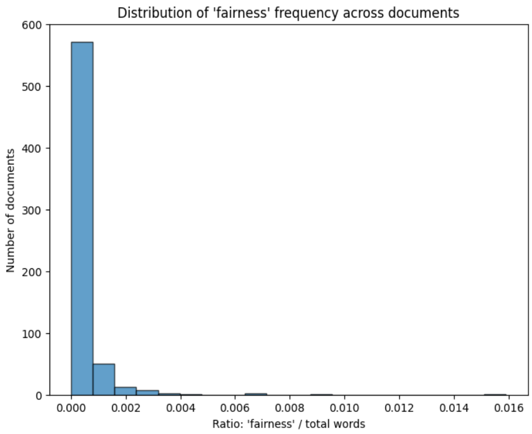
This scatter plot shows the relationship between the total number of words in a set of documents and the number of times the word "fairness" appears in each document.

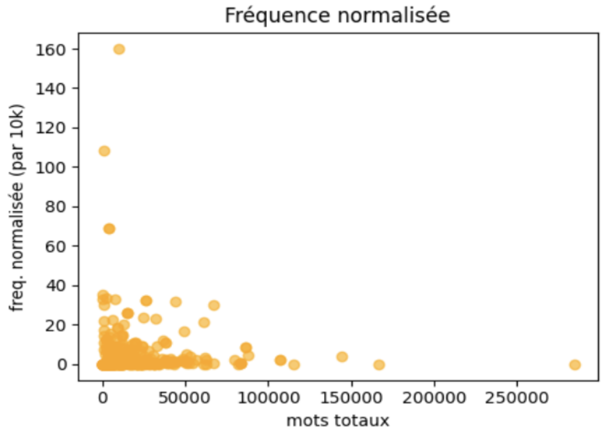
Most documents cluster in the lower-left corner, indicating both relatively short lengths and few occurrences of the term. However, a few outliers stand out with significantly higher counts of “fairness,” suggesting that some texts discuss the concept in much greater depth.

In short, raw counts mostly tell us where the volume is, not necessarily where the topic is central. A cluster of points is concentrated at the lower end of both axes, indicating that many documents are short and rarely mention "fairness".

The absence of a strong upward trend implies that there is no clear correlation between total word count and the number of times "fairness" is used.

We can conclude that the use of the term "fairness" appears to be independent of document length, suggesting **its usage may be topic-specific rather than a function of document size.**





The first histogram illustrates how the relative frequency of the word “fairness” is distributed across a collection of documents. Most documents have a very low ratio of “fairness” occurrences compared to their total word count. The vast majority cluster near zero on the x-axis. This indicates that, in most texts, “fairness” is mentioned extremely rarely or not at all. Only a handful of documents show higher ratios, suggesting that discussions focusing heavily on fairness are the exception rather than the rule.

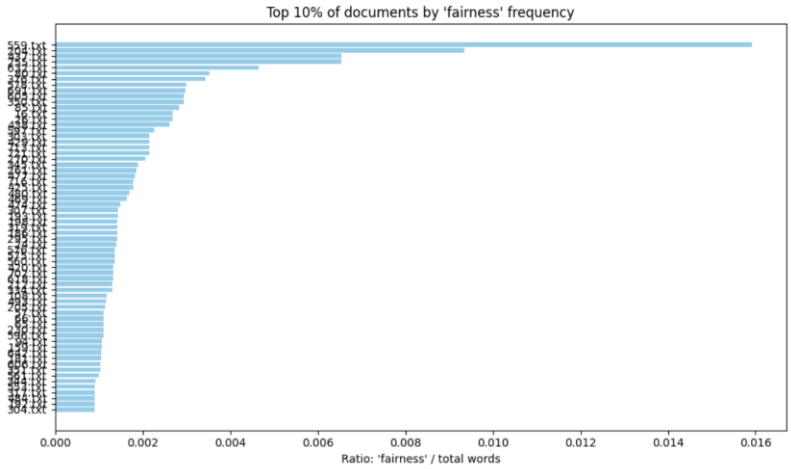
This kind of distribution is highly skewed to the right, a common pattern in linguistic data where a few documents contribute disproportionately to the use of a specific term. It implies that while “fairness” is a relevant topic, **it is not a dominant theme in the dataset as a whole.**

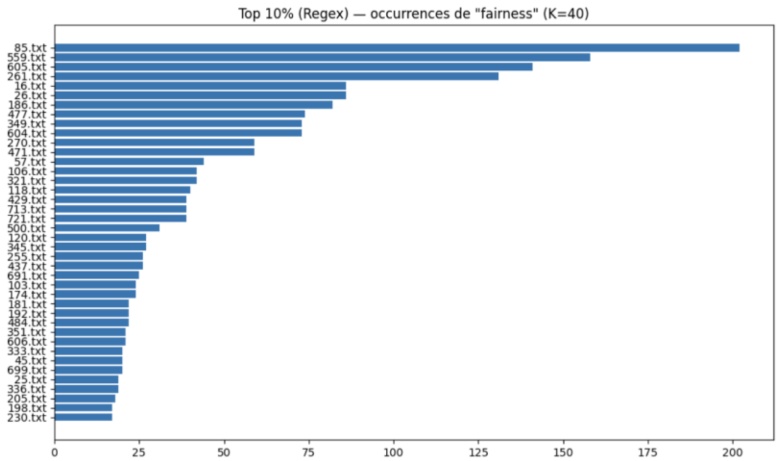
The second scatter plot shows the normalized frequency of “fairness” (per 10,000 words) against the total number of words in each document. Once again, most points lie close to the origin, revealing that documents with both a few and many words generally mention “fairness” at very low rates. However, some smaller documents exhibit unusually high normalized frequencies, indicating that “fairness” might be a central theme in a limited subset of shorter texts.

Together, these two figures suggest that while “fairness” appears sporadically across the corpus, its prominence varies widely. **A few documents seem to engage deeply with the topic, while the vast majority only touch upon it marginally or not at all.**

III.2. Documents in Which the Term Fairness Appears the Most

We rank documents by how much “fairness” they contain using two complementary pipelines: a fast regex tokenizer for a count-first baseline and an NLTK-based variant with word_tokenize plus English/French stopword removal for robustness. We also compute TF-IDF for the single term “fairness” (idf ≈ log((N+1)/(df+1))+1) to highlight documents where the word is unusually characteristic. We then select the top 10% by each criterion and visualize with horizontal bar charts (matplotlib). This dual view identifies both “high-volume” and “high-salience” sources for further close reading.





The two graphs present complementary perspectives on the distribution of the word **"fairness"** across a collection of text documents.

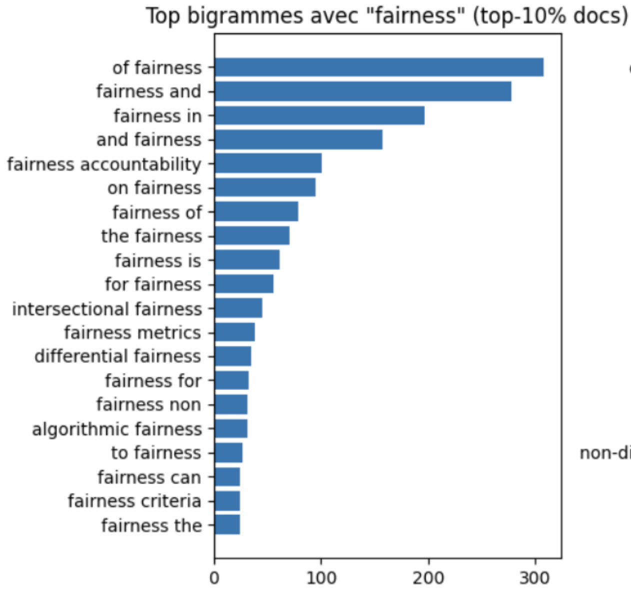
The first chart ranks the **top 10% of documents by the relative frequency** of "fairness". That is, the ratio of the word's occurrences to the total number of words in each document. This normalization highlights which texts discuss "fairness" most intensively, regardless of their length. Here, a few documents (like *559.txt* and *249.txt*) clearly stand out, suggesting that they dedicate a substantial portion of their content to this concept.

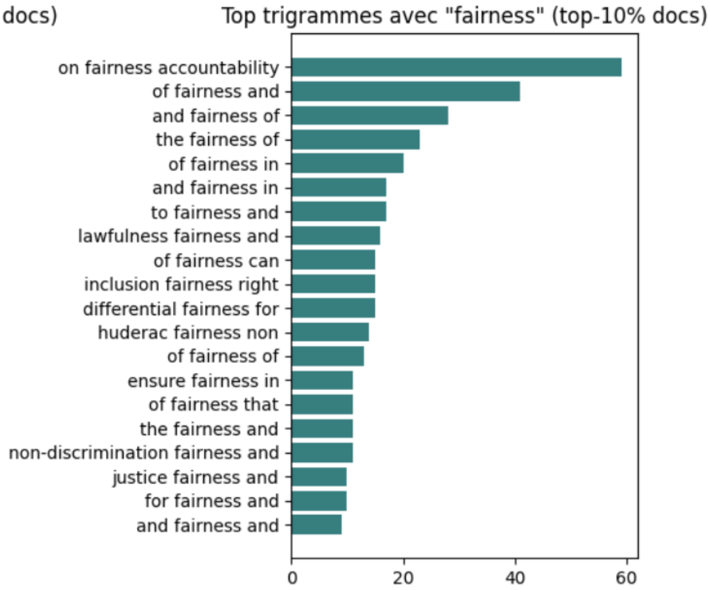
The second chart, by contrast, ranks the **top 10% of documents by absolute occurrences** of "fairness" detected via a regular expression. This view favors longer documents, since it counts total mentions without adjusting text size. Files such as *85.txt*, *559.txt*, and *605.txt* contain the highest raw counts, implying that these are either lengthy texts or ones where "fairness" is mentioned repeatedly throughout.

Together, these graphs show that while some documents contain many mentions of "fairness" simply because they are longer, others stand out because they emphasize the idea proportionally more.

III.3. Context in Which the Term Fairness Appears in the Documents

To probe local context, we extract windows around each "fairness" hit (regex tokenization), then aggregate bigrams and trigrams that include the target ("algorithmic fairness", "fairness in decision-making”, etc...). Frequency plots (matplotlib) surface the most common collocates and constructions, helping differentiate usage frames (policy, legal, technical).





The same story emerges: in the fairness-heavy slice of the corpus (top 10% of documents by the presence of the keyword), “fairness” most often appears as a noun that links to other concepts through simple prepositional or coordinating frames.

The bigrams are dominated by “of fairness”, “fairness and”, “fairness in”, and “and fairness”, with “of fairness” reaching roughly the 300 mark and the next two close behind. Linguistically, that means authors frequently evaluate “the fairness of” something, list fairness alongside neighboring principles (“fairness and …”), or situate it within a domain (“fairness in …”). Beneath these generic frames, the bigram list surfaces the main thematic pillars of the discourse: governance and accountability (fairness accountability, on fairness), operationalization and measurement (fairness metrics, fairness criteria), and domain-specific or advanced notions used in ML research (algorithmic fairness, intersectional fairness, differential fairness).

The trigram charts confirm and nuance these patterns. The most frequent sequence “on fairness accountability” points to a strong governance/policy current; “the fairness of” and “and fairness of” signal evaluative formulations (phrases or sentence structures that express evaluation or judgment) typical of methods sections or normative assessment; “ensure fairness in” and “of fairness in” indicate prescriptive guidance and application contexts; and expressions like “lawfulness fairness and”, “non-discrimination fairness” and “ inclusion fairness right” and “justice fairness and” reveal a legal-rights framing that often accompanies fairness discussions. Taken together, the collocations show a balanced mix **of conceptual/evaluative uses** (“the fairness of X”), **practical implementation and measurement** (“fairness metrics,” “ensure fairness in Y”), and **governance, accountability, and rights-based language**.

In short, the corpus talks about fairness as something to be measured and ensured in concrete systems, justified in normative terms, and anchored in accountability and non-discrimination frameworks.

III.4. Sentence-level analysis

We created a "fairness–MapAIE" sentence corpus using NLTK's `sent_tokenize` function (downloading the `punkt` model if it wasn't already available) and performed a simple case-insensitive search for the word *"fairness."* For each match, we recorded both the document ID—extracted from the filename using a regular expression—and the raw sentence text itself.

By working at the sentence level, however, we recognize that certain aspects are inevitably lost. First, we lose **discourse context**, including coreference links (who or what "it" or "this" refers to), cross-sentence cues such as "however" or "therefore," and the broader argumentative flow that extends beyond a single sentence. Second, we lose **document structure and metadata**, such as section or heading information, the author or source, and the document's type or provenance—all of which influence interpretation. Finally, we lose **broader semantic and pragmatic signals**, like negation or modality that span multiple sentences, definitions introduced earlier in the text, and examples that provide meaning across sentence boundaries.

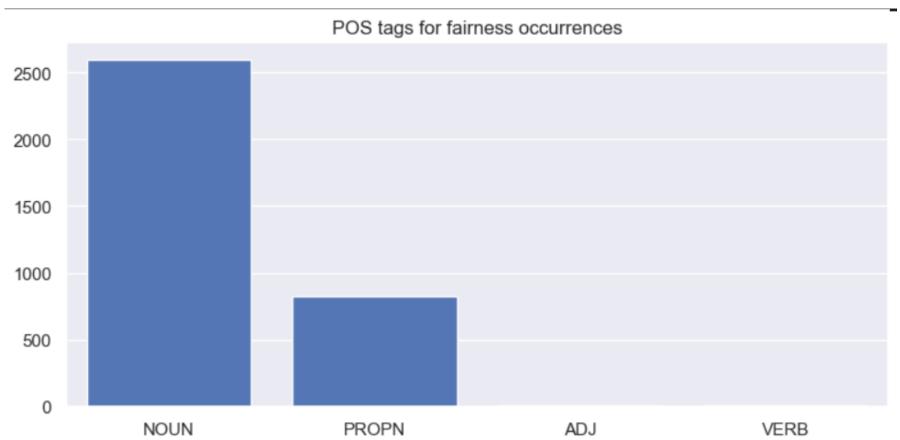
This sentence corpus will be used to produce data we need for AMR graphs exploration (see IV)

III.5. Syntactic roles taken by the term fairness in the corpus.

To further refine the linguistic characterization of the corpus, we examined the syntactic roles associated with the term fairness using POS (part-of-speech) tagging and dependency parsing.

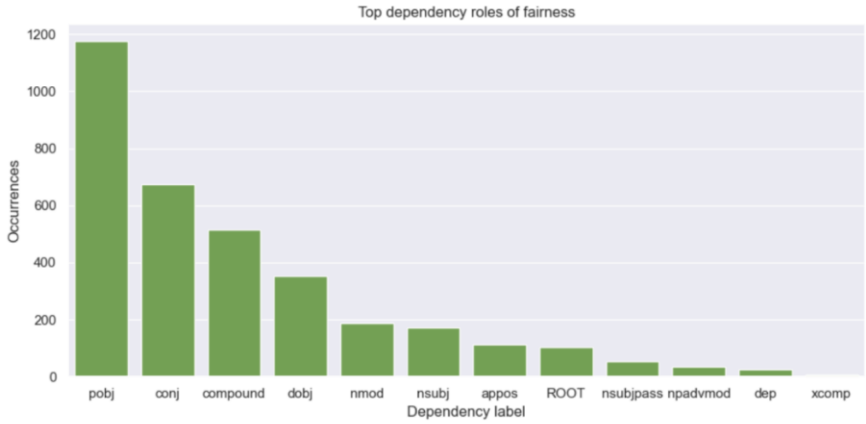
The goal of this analysis was to determine how fairness functions grammatically within sentences and, by extension, to infer the types of discursive structures in which it typically appears.

POS tagging results:



This graphic indicates that fairness overwhelmingly functions as a common noun (NOUN) with over 2500 occurrences in this category. This confirms that the term is primarily used as a conceptual entity, ethical principle, or value rather than an action. There is also a smaller proportion of occurrences classified as proper nouns (PROPN), reflecting instances where fairness appears capitalized (so "Fairness"), often in titles or section headings, or as a part of organizational names. Adjectival or verbal uses are negligible so the term is rarely used in descriptive forms in the corpus.

Dependency parsing results:



This graphic provides a more detailed view of the syntactic environments in which fairness appears. The most frequent dependency label is “pobj” (prepositional object), followed by “conj” (conjunct) and compound. It reveals that fairness frequently occurs as the object of prepositions such as “for”, “of”, “or”, “in”, for example, in expressions like “for fairness”, “principles of fairness”, etc.

Constructions like this position of fairness as the goal or thematic object of ethical reasoning.

The second most common dependency role “conjunct” indicates that fairness is often coordinated with other ethical concepts, probably like transparency, accountability, justice, etc.

The syntactic pattern supports the observation from the lexical analysis that fairness is embedded within clusters of related principles, forming part of enumerations of value.

Lastly, the presence of fairness as a compound (for example fairness assessment, fairness metrics...) highlights its frequent use in technical or evaluative contexts, where it modifies another noun to specify an ethical dimension of a process or evaluation.

IV. AMR Graphs Exploration

IV.1. Preliminary Study

Abstract Meaning Representation (AMR) is a framework that represents sentence meaning as a graph of concepts and relations. Below is an example of AMR:

```
# ::id 54
# ::snt Fairness requires that even careful decision processes provide some
avenue for redress of grievances.
(r / require-01
    :ARG0 (f / fairness)
    :ARG1 (p / provide-01
        :ARG0 (p2 / process-02
            :ARG1 (d / decide-01)
            :ARG1-of (c / care-04)
            :mod (e / even))
        :ARG1 (a / avenue
            :mod (s / some))
        :ARG2 (r2 / redress-01
            :ARG1 (c2 / complain-01))))
```

In this preliminary study on AMR, we focused on sentences containing the term fairness. From a total of 394 original text files, we extracted 7,043 sentences that include the term fairness. Among these, 500 representative sentences were selected as the research sample for AMR analysis, compiled into the dataset *fair_AMR-500.amr*.

The AMR graphs contain two concepts representing *fairness*: *fair-01* and *fairness*. While *fair-01* is defined in PropBank, fairness is not listed in the PropBank frames (https://propbank.github.io/v3.4.0/frames/). Consequently, we consider fairness and *fair-01* to be equivalent and unify them under a single conceptual category.

IV. 1.1. Preprocessing for the Multi-Sentence Problem

Due to the limitations of sentence splitting methods, certain texts were not correctly divided into multiple sentences. For example, the passage "*Considering ethical concepts such as justice, fairness, transparency, and accountability allows for valuable debate about the societal impacts of AI, and the role of AI in our lives.52 There is also an academic research community devoted to addressing ethical issues.*" was not successfully segmented into two sentences (see Table IV.1.1.1 - Orignal).

Original	**Preprocessed**

(m / multi-sentence
    :snt1 (a / allow-01
        ...
    :snt2 (d2 / devote-01
        ...
)
	(a / allow-01
	    ...
	)
	
	(d2 / devote-01
	    ...
	)

Table IV.1.1.1.

Although this issue does not pose a major problem for semantic interpretation, for the sake of analytical consistency we developed a preprocessing script to split multi-sentence blocks into separate units based on their AMR structure (see *multisentence.py* in our repository), treating each subordinate clause under multi-sentence as an independent AMR block (see Table III.1.1.1 - Preprocessed). Following this preprocessing step, we generated a refined set of 379 AMR graphs that include the fairness term. These graphs are compiled in the dataset *fair_AMR-500_clean.amr*.

IV. 1.2. AMR Exploration of the Term "Fairness"

AMR graphs provide access to an abstract representation of the semantic structure of the original sentences. In this analysis, we focus on the concept *fairness*, examining its semantic relations and its role within different context.

Figure IV.1.2.1 shows the AMR representation of the sentence "*Fairness requires that even careful decision processes provide some avenue for redress of grievances.*"

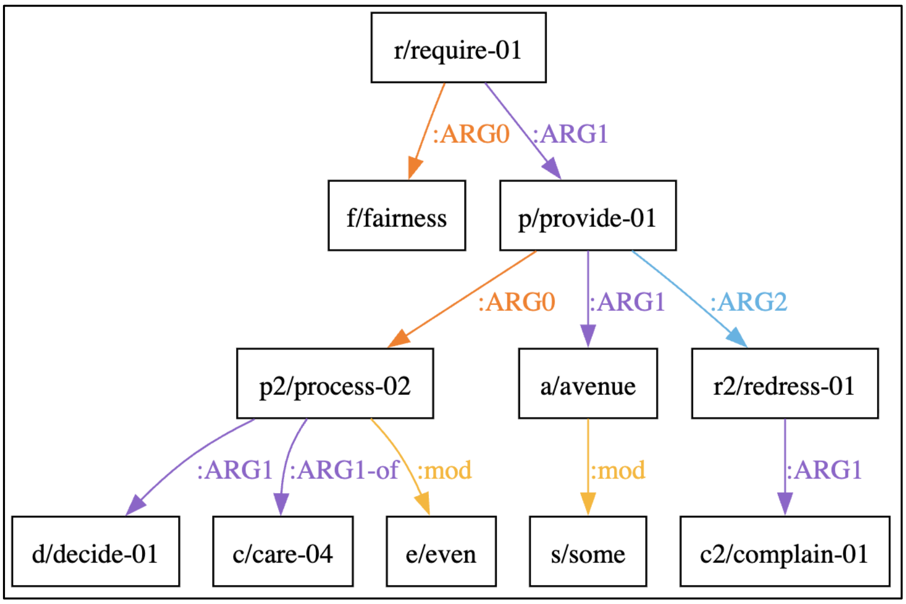


Figure IV.1.2.1. AMR graph rooted in r/require-01, where f/fairness functions as a core argument at an intermediate yet significant level. The graph below was rendered using the graphical editor Metamorphosed (https://github.com/Orange-OpenSource/metamorphosed).

In this AMR graph in Figure IV.1.2.1, the concept *f/fairness* occupies an intermediate but relatively important position. The root of the graph is *r/require-01*, and fairness functions as one of its core arguments.

- Parent: The direct parent of fairness is *require-01*, linked via the relation *:ARG0*, indicating that *fairness* is the agent or initiator of the requirement.
- Siblings: *Fairness* has one sibling, *p/provide-01*, connected to the same parent through *:ARG1*, representing what is required.
- Children: *Fairness* has no child nodes, making it a terminal concept within its local substructure.
- Relations: The only relation involving *fairness* is *:ARG0* (*require-01*, *fairness*), denoting that *fairness* performs the act of requiring.

Based on the *fair_AMR-500_clean.amr* dataset, we conducted a statistical analysis focusing on the following aspects:

- the **position** of the term *fairness* in the AMR graph (root / interior node / leaf);
- the **parents** (superordinate concepts) of *fairness*;
- the **siblings** (concepts sharing the same parent) of *fairness*;
- the **children** (subordinate concepts) of *fairness*; and
- the **relationships** linking *fairness* to its parents and children.

A summary of the results is presented below (by *analyze.py summary* in our repository):

```
=== Position of fairness ===
Position of fairness  count
                leaf    287
            interior     76
                root      6

=== Parent roles (relations) of fairness ===
Parent roles (relations) of fairness  count
                                  op    172
                                arg1     96
                               topic     27
                                 mod     25
                                arg2     19
                                arg0     16
                             purpose      9
                             example      3
                              domain      3
                                arg3      3
                                poss      1
                                part      1

=== Parent concepts of fairness ===
Parent concepts of fairness  count
                        and    165
                  define-01     16
                    mean-01     11
                     metric     10
                     notion      9
                   strategy      7
                   issue-02      6
                    formula      6
                approach-02      5
                     aspect      5
                         or      4
                     method      4
                  ensure-01      4
                       term      3
                 address-02      3
                 realize-01      3
                    concept      3
              multi-sentence      3
               determine-01      2
                 achieve-01      2
```

```
=== Child roles (relations) of fairness ===
Child roles (relations) of fairness  count
                                mod     50
                               arg1     14
                           polarity      7
                               poss      5
                             domain      4
                             manner      3
                               arg2      2
                                 li      2
                           location      2
                              topic      1
                        accompanier      1

=== Sibling concepts (same parent as fairness) ===
Sibling concepts (same parent as fairness)  count
                              transparency     65
                                   bias-01     34
                            accountable-02     30
                                       and     23
                                    ethics     16
                           discriminate-02     15
                                   privacy     14
                                     thing     11
                                   justice     10
                                  security      9
                           discriminate-01      8
                                   safe-01      8
                                explain-01      8
                                protect-01      8
                                  equal-01      8
                              impartiality      8
                                 diversity      6
                                  right-05      6
                             intelligent-01      6
                                    system      6
```

Position

Fairness most frequently appears as a leaf node (287 cases), occasionally as an interior node (76 cases), and rarely as a root (6 cases). This indicates that fairness typically functions as a referenced concept rather than an event predicate — it is something defined, modified, or exemplified, not something that governs other elements.

Parents

The dominant incoming role is *:op* (172 instances), showing that *fairness* is often part of enumerations or conceptual lists (e.g., "*justice, fairness, transparency…*"). Other common parent roles (relations) include *:ARG1, :topic*, and *:mod*, which link *fairness* to abstract predicates such as *define-01, mean-01, metric*, or *notion*. The most frequent parent concepts — *and, define-01, mean-01, metric*, and *notion* — confirm that *fairness* usually occurs in descriptive or definitional contexts rather than in action-oriented clauses.

Siblings
Typical co-listed siblings include *transparency, bias-01, accountable-02, ethics, justice*, and *privacy*. These strongly indicate that fairness belongs to a semantic cluster of ethical or normative principles, often enumerated together as values to be upheld or balanced (e.g., *fairness, accountability, transparency*).

Children
Outgoing roles are relatively sparse and shallow — the most frequent is *:mod* (50 instances), followed by *:ARG1* and *:polarity*, which attach small qualifiers rather than complex arguments. This supports the interpretation of fairness as a static abstract quality or moral attribute, not as a predicate introducing further events or relations.

Overall, *fairness* in AMR behaves primarily as a conceptual noun that appears in definitional or ethical contexts, rather than as something describing an action or process. Its syntactic and semantic position reflects its role as a value concept within the broader discourse on ethics and AI principles.

IV.2. Fairness Centrality Score

The previous section's analysis and data visualization are useful for illustrating the position and syntactic behavior of the fairness term in AMR graphs. However, to further quantify its importance within each sentence and to enable comparisons across different AMRs, we introduce a **Fairness Centrality Score**:

$$\text{Graph Score} = \max_{f \in \text{fairness nodes}} \left(\text{weight(role)} \times \frac{1}{1 + \text{distance(root, f)}} \right)$$

Where:

- distance(root,f) = shortest path length from the graph's root to the fairness node.
- weight(role) gives extra credit for important roles:
    - top node: 1.0,
    - :ARG0: 0.9, (relatively important)
    - :ARG0-of: 0.9, (relatively important)
    - :ARG1: 0.7,
    - :ARG2: 0.7,
    - :ARG3: 0.7,
    - :domain, :mod: 0.6
    - others (e.g., :op*): 0.4

Generally, graphs where fairness is positioned closer to the root and assumes core semantic roles tend to receive higher scores. For instance, the AMR graph in Figure IV.2.1 below obtains a score of 0.2 (relatively high).

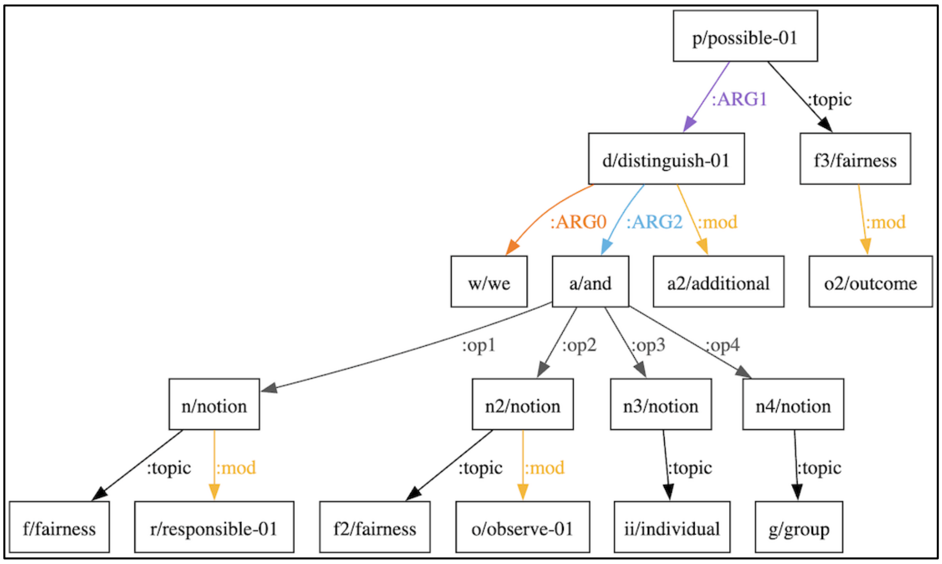


Figure IV.2.1. AMR graph of "*Within Outcome **Fairness** we can make additional distinctions, between Causal and Observational notions of **fairness**, as well as Individual and Group notions.*"

We calculated the Fairness Centrality Score for all graphs in *fair_AMR-500_clean.amr* using a script (*analyze.py centrality_score* in our repository). The histogram below shows the distribution of fairness centrality scores across the analyzed graphs. Most graphs have very low scores (below 0.2), indicating that *fairness* generally plays a minor role in the semantic structure of the corpus.

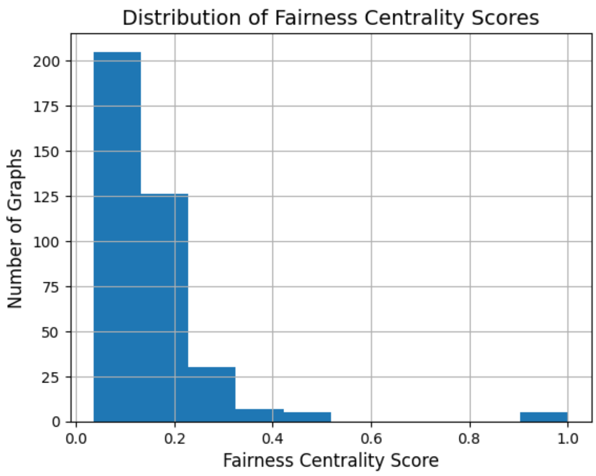


Table IV.2.1 below shows the top 10 sentences with the highest scores. Sentences with a score of 1.0 are those where fairness (or fair-01) is the root of the AMR, meaning the whole sentence centers on this concept. Lower scores (around 0.45) indicate that fairness appears in an important but non-root position. These sentences demonstrate direct, conceptual engagement with the notion of fairness rather than merely mentioning it in passing. Table IV.2.2 shows the three sentences with the lowest scores (score = 0.036), where fairness is only mentioned superficially alongside other ethical principles without further elaboration.

	gid	score	fairness_nodes	sentence
1	182	1.00	1	Algorithmic fairness
2	233	1.00	1	fairness
3	253	1.00	1	fairness
4	286	1.00	1	fairness
5	373	1.00	1	Group fairness between different schools
6	224	0.45	1	Fairness requires that even careful decision processes provide some avenue for redress of grievances.
7	314	0.45	1	With an increasing use of algorithmic decision-making systems in general, "fairness concerns ar e gaining momentum in academic and public discourses"
8	328	0.45	2	Procedural fairness is concerned with 'fair treatment' of people
9	329	0.45	1	Outcome fairness is concerned with what decisions are made
10	343	0.45	1	Fairness through unawareness is often not enough to prevent bias

Table IV.2.1. Top 10 sentences with the highest scores

	gid	score	fairness_nodes	sentence
376	263	0.036364	1	Besides, counterfactual explanations constitute a particularly interesting case of post -modelling explanatory method insofar as they can contribute to assessing that the appropriate data management principle described in this document has been followed, both in terms of regulatory compliance (specifically with respect to GDPR) and in terms of ethics and fairness.
377	265	0.036364	1	But it should be clear that the growing usage of AI must also strengthen societal robustness by building a ADDED ECONOMIC GROWTH, SCIENTIFIC EXCELLENCE AND HUMAN DEVELOPMENT clear vision of the impacts of AI in democracy, privacy, security, fairness, the labour market, governmental and commercial transparency, and equity.
378	272	0.036364	1	Important research lines are emerging, such as: Transparent AI: giving algorithms the ability to explain their own decisions and provide a high level and adaptive account of their workings to promote fairness and accountability; Emotional AI: algorithms will utilise emotions to achieve better decisions; Autonomous AI: important not only in the automotive sector but also in information systems, cybersecurity, smart cities, industry, etc.

Table IV.2.2. The three sentences with the lowest scores

"Fairness" rarely appears as an independent concept; instead, it functions as a contextual quality or attribute linked to broader notions — social or ethical — whose meaning depends on the surrounding system and purpose.

V. Conclusion

Our lexical investigation showed that fairness does not hold a central place within the broader ethical discourse, appearing alongside key notions such as data, ethics, and intelligence. Its frequency of use proved to be topic-dependent rather than proportional to document length: while a few texts engage with the term in depth, most only mention it in passing. Contextual analysis further revealed that fairness is typically framed in relation to governance, accountability, and rights-based perspectives, often appearing together with principles such as non-discrimination, justice, and accountability. From a syntactic standpoint, fairness most often functions as a common noun, serving either as the thematic object of ethical reasoning or as one item within a list of coordinated moral principles.

The AMR-based semantic analysis reinforced and quantified these observations. Across the 369 processed AMR graphs, fairness consistently appears as a conceptual or evaluative noun rather than an active process. It rarely serves as the main predicate (6 cases) but most frequently appears as a dependent node (287 cases) or as part of conceptual groupings via the :op role (172 instances), commonly associated with related terms such as transparency, bias, and privacy. The introduction of the Fairness Centrality Score helped measure its semantic prominence within sentences: while some sentences revolve around fairness, in most, it plays a more supportive, contextual role.

In sum, the corpus portrays fairness not as an independent action but as a foundational ethical value, one that must be assessed, safeguarded within concrete systems, and grounded in normative frameworks. This linguistic profile underscores the ongoing tension between the abstract principle of fairness and the practical challenges of embedding it into AI governance.